



Mark Scheme (Results)

January 2025

Pearson BTEC Nationals in
Applied Science/Forensic and Criminal
Investigation (31617H/1C)
Unit 1: Principles and Applications of Science I
(Chemistry)

Edexcel and BTEC Qualifications

Edexcel and BTEC qualifications come from Pearson, the world's leading learning company. We provide a wide range of qualifications including academic, vocational, occupational and specific programmes for employers. For further information visit our qualifications website at <http://qualifications.pearson.com/en/home.html> for our BTEC qualifications.

Alternatively, you can get in touch with us using the details on our contact us page at <http://qualifications.pearson.com/en/contact-us.html>

If you have any subject specific questions about this specification that require the help of a subject specialist, you can speak directly to the subject team at Pearson. Their contact details can be found on this at <http://qualifications.pearson.com/en/support/support-for-you/teachers.html>

You can also use our online Ask the Expert service at <https://www.edexcelonline.com>. You will need an Edexcel Online username and password to access this service.

Pearson: helping people progress, everywhere

Our aim is to help everyone progress in their lives through education. We believe in every kind of learning, for all kinds of people, wherever they are in the world. We've been involved in education for over 150 years, and by working across 70 countries, in 100 languages, we have built an international reputation for our commitment to high standards and raising achievement through innovation in education. Find out more about how we can help you and your learners at: <https://www.pearson.com/uk>

January 2025

Publications Code 31617H/1C_MS_2501

All the material in this publication is copyright

© Pearson Education Ltd 2025

Unit 1: Principles and Applications of Science I (Chemistry)

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

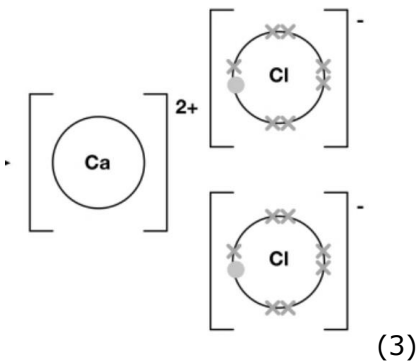
Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Additional Guidance	Mark
1(a)	B 18.0		1
1(b)	covalent		1
1(c)	D 1, 2 and 3		1
1(d)	any two from: {hydrogen / H₂ / H} has fewer electron(s) (than oxygen) (1) <u>weaker</u> Van der Waals forces (1) therefore less {energy/heat} to break (Van der Waals forces) (1)	accept ORA accept hydrogen <u>only</u> has 1 electron accept hydrogen molecules have 2 electrons whereas oxygen molecules have 16 electrons ignore references to protons and neutrons accept <u>weaker</u> {London forces / temporary dipole forces / intermolecular forces / intermolecular bonds / forces between molecules} accept London forces / temporary dipole forces / intermolecular forces / intermolecular bonds / forces between molecules for Van der Waals ignore easier to break ignore less energy to break bonds alone reject less energy to break hydrogen bonds	2
Total = 5 marks			

Question Number	Answer	Additional Guidance	Mark
2(a)(i)	effervescence / fizzing / bubbling / test tube feels warmer	accept magnesium disappears accept magnesium {moves / floats} ignore magnesium {reacts / dissolves} ignore {hydrogen / gas} is formed ignore temperature increase reject incorrect observations such as flame	1
2(a)(ii)	Mg + 2HCl → MgCl ₂ + H ₂ (2) OR all correct formula (1) balancing of correct formula (1)	accept correct multiples ignore state symbols reject non-subscripts and incorrect cases in formula	2
2(b)(i)	1.825 ÷ 36.5 = 0.05 (1)	accept full marks for correct answer of 0.05 without working	1
2(b)(ii)	using answer of 0.05 from 2(b)(i) conversion of volume (1) 25 ÷ 1000 = 0.025 calculation of concentration (1) (0.05) ÷ 0.025 evaluation (1) 2 (mol dm ⁻³) OR Using value 0.07 conversion of volume (1) 25 ÷ 1000 = 0.025 calculation of concentration (1) 0.07 ÷ 0.025 evaluation (1) 2.8 (mol dm ⁻³)	accept ecf from 2(b)(i) accept full marks for correct answer of 2 without working accept ecf accept ecf	3

Question Number	Answer	Additional Guidance	Mark
2(c)	reaction would be more vigorous (with sodium) (1) sodium only needs <u>to lose one</u> electron (1)	accept ORA accept sodium is more reactive (than magnesium) accept reaction is faster with sodium accept descriptions of a more vigorous reaction in terms of correct observations accept sodium <u>needs to lose</u> fewer electrons accept sodium <u>loses its electron</u> more easily ignore sodium has one electron on its outer shell alone	2
Total = 9 marks			

Question Number	Answer	Additional Guidance	Mark
3(a)	C 4s		1
3(b)(i)	any two from: high {melting / boiling} point (1) conducts electricity when {in solution / molten / in liquid state} (1) soluble (in water) (1)	ignore any chemical properties given do not accept higher {melting / boiling} point accept does not conduct (when solid) accept brittle accept white colour	2
3(b)(ii)	 OR electron arrangement with charge for calcium (1) electron arrangement with charge for one chloride ion (1) electron arrangement with charge for a second chloride ion (1)	ignore inner shells correct or otherwise any sharing of electrons scores 0 accept calcium with eight electrons if electron configurations are correct but there is incorrect use of dots and crosses, allow 1 mark for correct charges on all ions.	3

Question Number	Answer	Additional Guidance	Mark
3(b)(iii)	An explanation linking: calcium has {been oxidised / lost electrons} oxidation number of calcium has changed from 0 to +2 chlorine has {been reduced / gained an electron} oxidation number of chlorine has changed from 0 to -1	accept chlorine has gained electrons reject oxidation number of chlorine has changed from 0 to -2	4
Total = 10 marks			

Question number	Indicative Content
4	<i>Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and levels descriptors below. The indicative content that follows is not prescriptive. Answers may cover some or all of the indicative content, but learners should be rewarded for other relevant answers.</i> • electronegativity is the tendency of an atom to attract a bonding pair of electrons in a covalent bond groups • electronegativity decreases going down groups • number of electron shells increases down the groups • number of protons increases down the group • atomic radius increases down the group • weaker attraction of electron pair to nucleus • due to shielding periods • electronegativity increases across the period • same number of shells • same shielding • more protons in the nucleus • atomic radius decreases across the group • therefore stronger attraction of electron pair to nucleus noble gases • no electronegativity value for He, Ne and Ar because noble gases have full outermost shells • do not form bonds • electron pair pull cannot be determined.

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Adequate interpretation, analysis and / or evaluation of the scientific information with generalised comments being made. • Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported. • The discussion shows some structure and coherence.
Level 2	3–4	• Good analysis, interpretation and / or evaluation of the scientific information. • Lines of argument mostly supported through the application of relevant evidence. • The discussion shows a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Comprehensive analysis, interpretation and / or evaluation of all pieces of scientific information. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence. • The discussion shows a well-developed structure which is clear, coherent and logical.
Total = 6 marks		



Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

